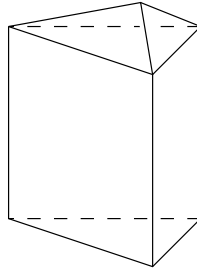


Problems and Exercises

- 5.0 Prove that every polytopal subdivision of a polygon without new vertices is regular.
- 5.1 Show that the following 3-polytope (the *capped prism*) has a nonregular triangulation without new vertices.
(Kleinschmidt and Lee, see Lee [357, Sect. 6])



- 5.2 Let $\mathcal{C}$ be any polytopal subdivision of a d -polytope in $\mathbb{R}^d$.
- (i) Show that the decision of whether $\mathcal{C}$ is regular can be reduced to a linear programming feasibility problem. How do you get rid of the strict inequalities that come up?
 - (ii) If $\mathcal{C}$ is a Schlegel diagram, show that it can be obtained from a “lower faces” construction as in Definition 5.3.
(Use a projective transformation that moves $\mathbf{y}_F$ “to infinity”; compare to Exercise 2.18.)
 - (iii) If $\mathcal{C}$ is a d -diagram, show that the question of whether it is a Schlegel diagram can also be reduced to a linear programming feasibility problem.
- 5.3 Produce examples of polytopes that have nonregular triangulations without new vertices.
- (i) The 4-cube C_4 has nonregular triangulations without new vertices.
(It was an open problem for a long time whether C_d has nonregular triangulations for any d . Now De Loera [182] has shown that there are even such triangulations with 24 maximal simplices, all of which have volume $1/4!$.)
 - (ii) The second hypersimplices $\Delta_{d-1}(2)$ have nonregular triangulations without new vertices, for $d \geq 9$.
(These were constructed with combinatorial tools by De Loera, Sturmfels & Thomas [183]; you are allowed to use the computer, via part (i) of the previous exercise. This might in particular be useful in the next two parts, which are unsolved problems.)

- (iii) Can products $\Delta_{m-1} \times \Delta_{n-1}$ of two simplices have nonregular triangulations?

(Answer: $\Delta_3 \times \Delta_3$ is the product to look at – De Loera [182]).

- (iv)* Consider the triangulations of the square $[0, n]^2$ with vertex set $\{0, 1, \dots, n\} \times \{0, 1, \dots, n\}$. Estimate the numbers $f(n)$ and $f_{\text{reg}}(n)$ of all triangulations resp. of all regular triangulations, as precisely as possible. (Compare your results to those in [298].) Is it true that “most of” the triangulations are non-regular, when n gets large, that is, that the ratio $f_{\text{reg}}(n)/f(n)$ tends to 0 for $n \rightarrow \infty$?

- 5.4 Define the product of two polyhedral complexes, in such a way that the product of subdivisions of two polytopes P and Q is a subdivision of the product $P \times Q$.

Prove that the product subdivision is regular if and only if the original subdivisions of P and Q were regular.

- 5.5 Compute the numbers f_k of k -faces for the polytope $\tilde{\mathcal{P}}_4(z_1, z_2, z_3)$.

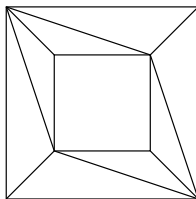
- 5.6 A “default” convex function is given by the paraboloid function

$$f: \mathbb{R}^d \longrightarrow \mathbb{R}, \quad \mathbf{x} \mapsto \sum_{i=1}^d x_i^2.$$

Let $V \subseteq \mathbb{R}^d$ be a finite set of points (vertices). Show that the regular subdivision of $Q := \text{conv}(V)$ associated with this f has the following property: for every facet of the subdivision there is a sphere that contains all its vertices, but no other vertices from V .

(This subdivision is known as the *Delaunay triangulation* of the point set V . It is of great importance for many aspects of computational geometry; see for example Edelsbrunner [190].)

- 5.7 Show that the following 2-diagram is regular, but not Schlegel:



- 5.8 Show that in general the polytopal complex $\mathcal{C}(\partial P) \setminus \{F\}$ and the Schlegel diagram $\mathcal{D}(P, F)$ are not affinely isomorphic. (For this, it suffices to consider the 3-cube!)

However, show that $p: G \rightarrow p(G)$ is a projective transformation, for all proper faces G of P .

5.9 Construct a Schlegel diagram for the polar of a product

$$P = (\Delta_2 \times \Delta_2)^\Delta.$$

Construct Schlegel diagrams for the cyclic polytopes $C_3(6)$ and $C_4(7)$.

5.10* What is the smallest number $f(d)$ of d -simplices that is sufficient to triangulate a d -cube?

Combining results by Hughes and Anderson [285] [287] [286] [288] and Haiman [267], we know that

$$f(2) = 2 \quad f(3) = 5 \quad f(4) = 16 \quad f(5) = 67$$

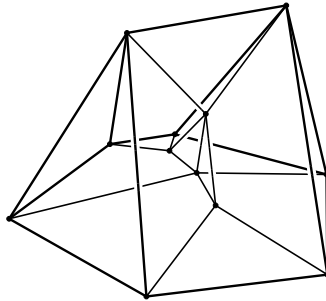
$$f(6) = 308 \quad f(7) = 1493 \quad \text{and} \quad 5522 \leq f(8) \leq 11944.$$

For large d a method due to Smith [505], using volume estimates in hyperbolic geometry, yields the best lower bounds on $f(d)$ so far.

Is it true that the smallest number is always/only achieved by a triangulation without new vertices?

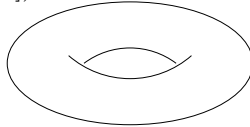
What is the maximal number of simplices that may be needed to triangulate *any* 0/1-polytope: Is this the same number you get in case of the d -cube?

(“Efficient” triangulations of d -cubes, with few facets, have been studied extensively, for example for use in finite element methods for solving differential equations. Thus, a lot is known about such triangulations. For example, one knows that asymptotically, at most $\rho^d d!$ simplices are needed for large d , for some constant $\rho < 1$. We refer to Haiman [267] and Todd & Tunçel [545] for more information.)

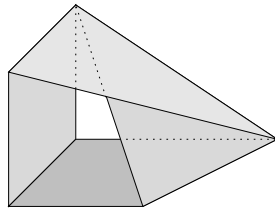
5.11 For a 4-polytope, one cannot prescribe the shape of a 2-face! Namely, Richter-Gebert [459, p. 91] [462] provides the following diagram of a 4-dimensional polytope X^* :

Show that this represents the Schlegel diagram of a 4-polytope with 8 facets and 12 vertices. Show that the shape of the hexagon at the base cannot be prescribed arbitrarily: three lines, determined by two opposite edges and by the diagonal between them, must (projectively) meet in a point. (See also Exercise 6.11)

- 5.12 Show that 2^d simplices can be arranged in $\mathbb{R}^d$ in such a way that any two are adjacent (that is, the intersections are $(d - 1)$ -dimensional).
- 5.13 The smallest triangulation of the torus surface has 7 vertices, 21 edges and 14 triangles. Construct it, and show that it can be embedded as a subcomplex into $\mathcal{C}(C_7(4))$, and thus as a simplicial complex into $\mathbb{R}^3$. (Császár [169], Grünbaum [252, p. 253]; see also Altshuler [14] and Bokowski & Eggert [113])



- 5.14* Can *every* triangulation of the torus be realized by a simplicial complex $\mathcal{C}$ in $\mathbb{R}^3$? (This is a classical problem of Grünbaum [252, p. 253], which is still wide open. See Ewald, Kleinschmidt, Pachner & Schulz [203, p. 153], Altshuler, Bokowski & Schuchert [15] and their references.)
- 5.15 Show that every d -dimensional simplicial complex can be realized as a subcomplex of a simplicial $(2d+2)$ -polytope, and thus has a straight realization as a simplicial complex in $\mathbb{R}^{2d+1}$. (Hint: Take a suitable cyclic polytope and its Schlegel diagram.)
- 5.16 Show that there are polyhedral complexes that are not subcomplexes of polytopes. Namely, show that the subdivision of the *Möbius band* drawn below with 6 vertices, 10 edges, and 4 facets can be realized as a polyhedral complex in $\mathbb{R}^3$, but not as a subcomplex of any polytope.



(This is due to Betke, Schulz & Wills [68]; see Barnette [48] for a similar, but simplicial, Möbius strip that serves as an “impediment for polyhedrality.” It is not even true that every triangulated Möbius strip has a straight embedding into $\mathbb{R}^3$: see Brehm [129]. However, the Schlegel diagram construction shows that no such Möbius strip can appear in the boundary of a 4-polytope.)

- 5.17 Modify Schulz’ 3-diagram $\mathcal{D}_1$ from Example 5.10 by subdividing the bipyramid A into three tetrahedra, $[2, 3, 6]$, $[2, 5, 6]$, and $[2, 6, 7]$. Show that this yields a new 3-diagram $\mathcal{D}'_1$ with 8 vertices and 12 facets, which contains the Möbius band of the previous exercise as a subcomplex. Derive that $\mathcal{D}'_1$ is not polytopal, either.

6

Duality, Gale Diagrams, and Applications

More about life in high dimensions: after “successfully” dealing with polytopes in four dimensions, we now study polytopes with few vertices, that is, d -polytopes with only d -plus-a-few vertices. For this, we develop a duality theory, which describes them in terms of structures in low dimensions.

This duality theory (developed by Perles in the 1960s, and recorded by Grünbaum [252]) is classically known as *Gale diagrams*. Later it was realized (apparently first by McMullen [394]) that Gale diagrams are a manifestation of “oriented matroid duality.”

Thus behind the construction of Gale diagrams one finds (barely hidden) *oriented matroids*. Their theory was initiated in the 1970s by at least four independent authors, Jon Folkman, Jim Lawrence, Robert G. Bland, and Michel Las Vergnas; see [207] and [100].

By now oriented matroids form a theory with many facets, extensive enough to fill thick books [96]. One aim of our lectures is to give a simple introduction to a few topics that help understand polytopes. Keys to this include the description of the relative position of vertices in terms of “circuits” and “cocircuits,” as well as the simple duality between these two descriptions, *oriented matroid duality*, which in polytope theory manifests itself in (linear and affine) Gale diagrams.

Thus this lecture includes a brief crash (?) course on oriented matroids. (This will be continued in Lecture 7, when we discuss hyperplane arrangements and zonotopes.) Although oriented matroids need some amount of new notation and terminology, there is little magic involved: just don’t be scared of names. For further reading, we refer to the “Orientation Session” in Björner et al. [96, Ch. 1].